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MULTIOMICS DATA SCIENCE - ENRICHMENT ANALYSIS

ENRICHMENT ANALYSIS

- Statistical concept to check a set of biomolecules is associated to a certain state like:
  - Biological function
  - Metabolic pathway
  - Disease
- Known methods:
  - Over-representation analysis (ORA)
  - Gene set enrichment analysis (GSEA)

OVER-REPRESENTATION ANALYSIS

- Statistical method to check if entries of a pre-defined subset are over-represented in a subset of own data

- Other point of view: is there a correlation between the feature forming the first subset and the second feature forming the second subset

OVER-REPRESENTATION ANALYSIS: EXAMPLE

- Let‘s assume we analyze a set of 100 people
- Some of them are students (say 23 students), some don‘t
- Some of them read books in their spare time (say 43 book readers), some don‘t

OVER-REPRESENTATION ANALYSIS: EXAMPLE

- Let‘s assume we analyze a set of 100 people
- Some of them are students (say 23 students), some don‘t
- Some of them read books in their spare time (say 43 book readers), some don‘t
- Question: is there a correlation between being a student and reading books in the spare time?
- Null hypothesis: there is no correlation between being a student and reading books in the spare time

OVER-REPRESENTATION ANALYSIS: EXAMPLE

	student	*no student*
read books	17	26
no books	6	51

OVER-REPRESENTATION ANALYSIS: EXAMPLE

	student	*no student*	*total*
read books	17	26	43
no books	6	51	57
total	23	77	100

FISHER EXACT TEST

- Test for analyses on contingency tables
- Exact calculation of p-value
- Copes well (statistically stable) also for small sample sizes

FISHER EXACT TEST: EXAMPLE

	student	*no student*	*total*
read books	A	B	A+B
no books	C	D	C+D
total	A+C	B+D	N

- Hypergeometric distribution of numbers in the cells
- p-value: $p = \frac{\binom{A+B}{A} \cdot \binom{C+D}{C}}{\binom{N}{A+B}}$
- Probability of observing this particular arrangement of the data

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FISHER EXACT TEST: EXAMPLE

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- Hypergeometric distribution of numbers in the cells
- p-value: $p = \frac{\binom{A+B}{A} \cdot \binom{C+D}{C}}{\binom{N}{A+B}}$
- Probability of observing this particular arrangement of the data
- For the students / books example: p = 0.00061465581
- But, do we want this probability???

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FISHER EXACT TEST: EXAMPLE

	student	*no student*	*total*
read books	17	26	43
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total	23	77	100

- Actually we want a total probability of all situations that are as extreme or more extreme than our current observation
- We take the p-value as a measure of extremeness
- What are the other situations?

FISHER EXACT TEST: EXAMPLE

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- Actually we want a total probability of all situations that are as extreme or more extreme than our current observation
- We take the p-value as a measure of extremeness
- What are the other situations?
- Examples:

	student	*no student*	*total*
read books	12	31	43
no books	11	46	57
total	23	77	100

	student	*no student*	*total*
read books	1	42	43
no books	22	35	57
total	23	77	100

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FISHER EXACT TEST: EXAMPLE

	student	*no student*	*total*
read books	17	26	43
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FISHER EXACT TEST: EXAMPLE

	student	*no student*	*total*
read books	17	26	43
no books	6	51	57
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$p = 6.146 \cdot 10^{-4}$

- Actually we want a total probability of all situations that are as extreme or more extreme than our current observation
- We take the p-value as a measure of extremeness
- What are the other situations?
- Examples:

	student	*no student*	*total*
read books	12	31	43
no books	11	46	57
total	23	77	100

$p = 0.113$

	student	*no student*	*total*
read books	1	42	43
no books	22	35	57
total	23	77	100

$p = 6.034 \cdot 10^{-6}$

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COMPUTE SITUATIONS

For *i* between 0 and min(# students, # book readers) included:

	student	*no student*	*total*
read books	*i*	43 - *i*	43
no books	23 - *i*	100 + i - 23 - 43	57
total	23	77	100

```
from scipy.stats import hypergeom as hyp
def hgp(a, b, c, d): return hyp.pmf(a, a + b + c + d, a + b, a + c)

def enrichment(a, b, c, d):
    n, B, S, p_cut = a + b + c + d, a + b, a + c, hgp(a, b, c, d)
    p = [hgp(i, S - i, B - i, n + i - S - B) for i in range(min(S, B) + 1)]
    return sum([pv for pv in p if pv <= p_cut])
```

COMPUTE SITUATIONS

For *i* between 0 and min(# students, # book readers) included:

	student	*no student*	*total*
read books	*i*	43 - *i*	43
no books	23 - *i*	100 + i - 23 - 43	57
total	23	77	100

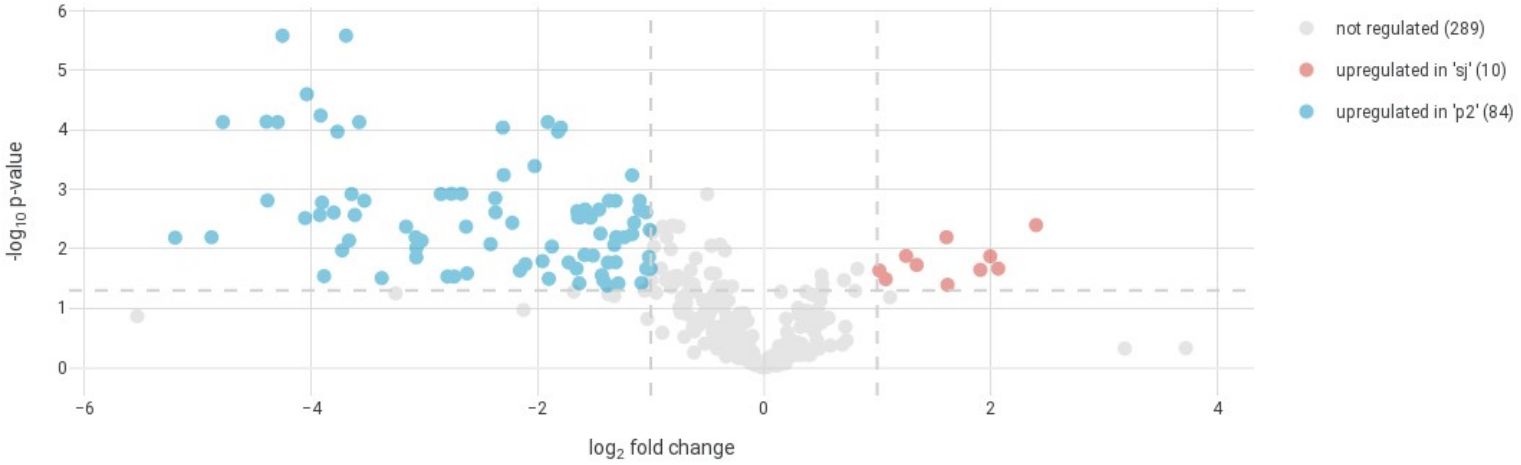
```
from scipy.stats import fisher_exact

def enrichment(a, b, c, d):
    # does exactly the same, but faster and quicker
    return fisher_exact(table = [[a, b],[c, d]]).pvalue
```

POSSIBLE SUBSETS OF BIOMOLECULES

- Up/down regulation of biomolecules within conditions in (study / clinical) parameters
- Biomolecules in certain pathways
- Proteins associated with in gene ontology terms

UP/DOWN REGULATION OF BIOMOLECULES (VOLCANO PLOTS)

- Take a molecule and compare all its measured values within one condition against another condition (e.g., healthy vs. sick), in terms of:
  - p-value of statistical test
  - fold change: avg(condition$_1$) / avg(condition$_2$)



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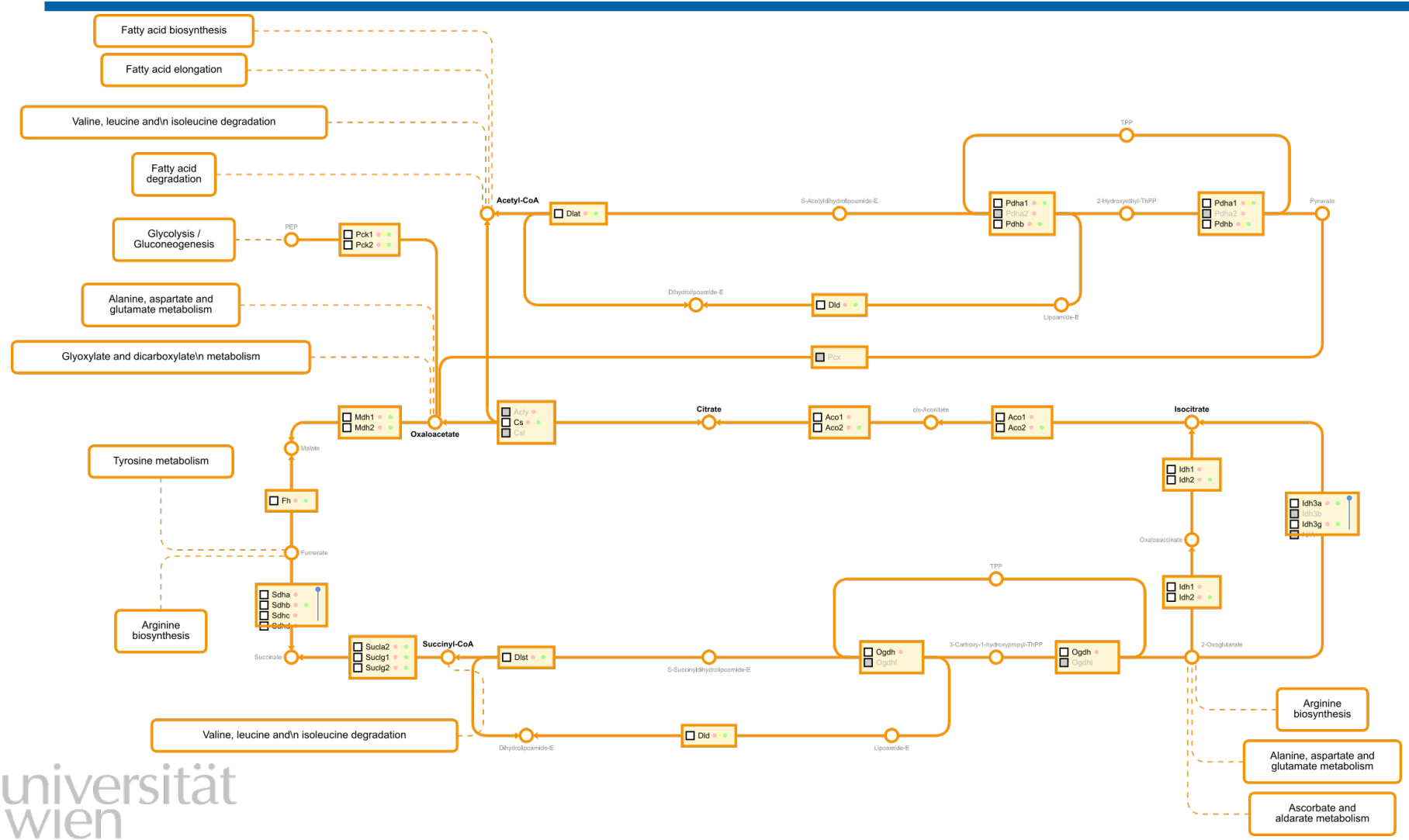
METABOLIC PATHWAYS

- Metabolic pathways describe enzymatic reactions of metabolites and biomolecules

- Each pathway has:
  - a certain function
  - a set of associated proteins (enzymes)
  - a set of associated metabolites / lipids



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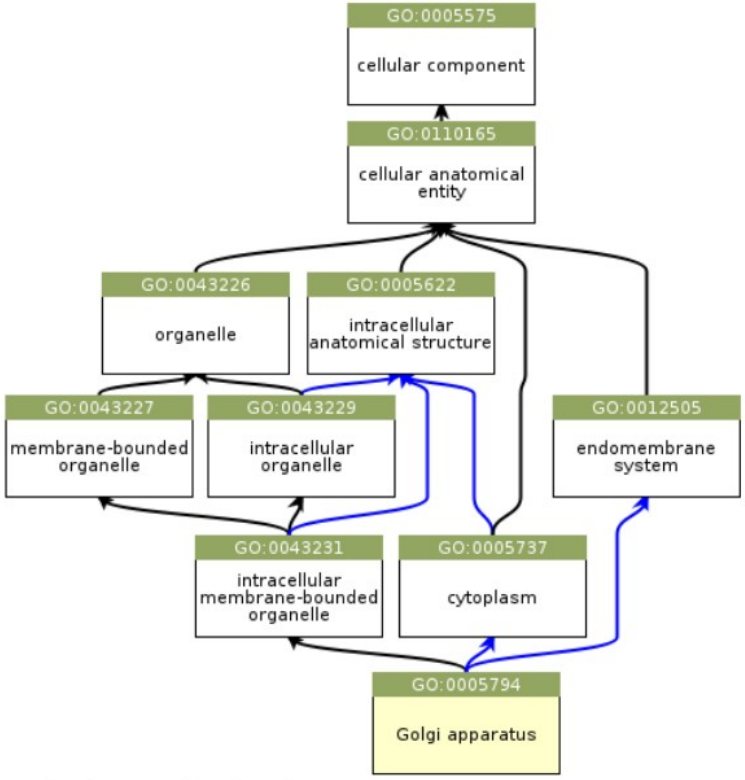
GENE ONTOLOGY

- Hierarchically structures set of terms within three domains:
  - Biological processes
  - Cellular components
  - Molecular functions
- Proteins are associated to some of these terms:
- Example:
  - Sonic hedgehog protein (Q15465)
    - ↳ Golgi apparatus (GO:0005794)



QuickGO - https://www.ebi.ac.uk/QuickGO

ENRICHMENT ANALYSIS - EXAMPLE

- Assumption: oxidative stress has an effect on the golgi apparatus (GA)
- Null hypothesis: nope, it doesn‘t

ENRICHMENT ANALYSIS - EXAMPLE

- Assumption: oxidative stress has an effect on the golgi apparatus (GA)
- Null hypothesis: nope, it doesn‘t
- Step 1) measure your biomolecules (say proteins) on a control cohort and a cohort suffering from oxidative stress
  - you obtain n proteins over $m_C + m_{OS}$ cohorts / observations
- Step 2) determine which of the n proteins are regulated for oxidative stress (A + B)

ENRICHMENT ANALYSIS - EXAMPLE

- Step 3) count how many of the *n* proteins are associated with the GA (A + C)
- Step 4) count how many proteins are regulated and within the GA (A)

ENRICHMENT ANALYSIS - EXAMPLE

- Step 3) count how many of the *n* proteins are associated with the GA (A + C)
- Step 4) count how many proteins are regulated and within the GA (A)
- Step 5) from (A), (A + B), (A + C), and *n* determine B, C, D
- Step 6) compute the enrichment probability p

ENRICHMENT ANALYSIS - SECOND EXAMPLE

- Explorative approach: which pathways are regulated during oxidative stress?
- Step 1) take your experiment from before with n proteins over m_C + m_{OS} cohorts / observations
- Step 2) determine which of the n proteins are regulated for oxidative stress (A + B)

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ENRICHMENT ANALYSIS - SECOND EXAMPLE

- Explorative approach: which pathways are regulated during oxidative stress?
- Step 1) take your experiment from before with n proteins over $m_C + m_{OS}$ cohorts / observations
- Step 2) determine which of the n proteins are regulated for oxidative stress (A + B)
- Step 3) select your k pathways of interest
- Step 4) for each $\text{pathway}_{i \in k}$: count how many proteins are in pathway_i (A + C)

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ENRICHMENT ANALYSIS - SECOND EXAMPLE

- Step 5) count how many regulated proteins are in $pathway_i$ (A)
- Step 6) from (A), (A + B), (A + C), and *n* determine B, C, D

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ENRICHMENT ANALYSIS - SECOND EXAMPLE

- Step 5) count how many regulated proteins are in pathway$_i$ (A)
- Step 6) from (A), (A + B), (A + C), and *n* determine B, C, D
- Step 7) compute the enrichment probability p_i
- Step 8) compute all *k* enrichment probabilities

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ENRICHMENT ANALYSIS - SECOND EXAMPLE

- Step 5) count how many regulated proteins are in pathway$_i$ (A)
- Step 6) from (A), (A + B), (A + C), and *n* determine B, C, D
- Step 7) compute the enrichment probability p_i
- Step 8) compute all *k* enrichment probabilities
- Step 9) correct all p values due to the multiple testing approach
- Step 10) determine which pathways are still (very/highly) significant

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SUMMARY

- This course introduced the following topics on data analysis:
  - regression
  - principal component analysis
  - clustering / time series analysis
  - classification
  - statistical testing
  - enrichment analysis
- And a crash course in Python with creating figures for data science

Quelle: *https://cdn.shopify.com/s/files/1/0268/8219/products/dont_panic.jpg?v=1571264365*